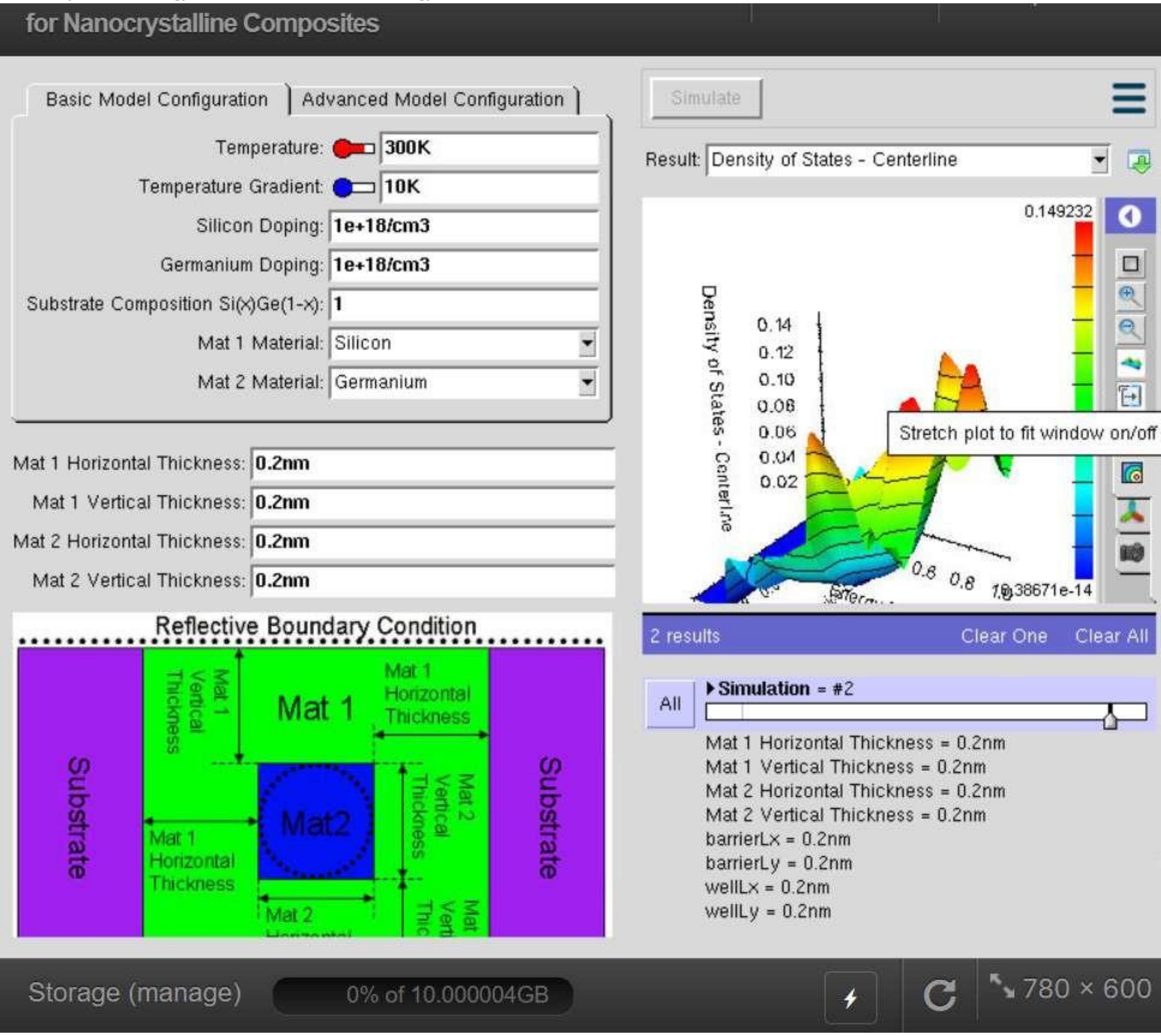
Course code: ECA3102 Course: Nano electronics Exp 1: Simulation of Density of States of Quantum well structure with various nano meter thickness.

Why ? The purpose of this projecTt is to understand how **quantum confinement** changes the **Density of States (DOS)** when the thickness of a quantum well is varied from a few nanometers to several nanometers.

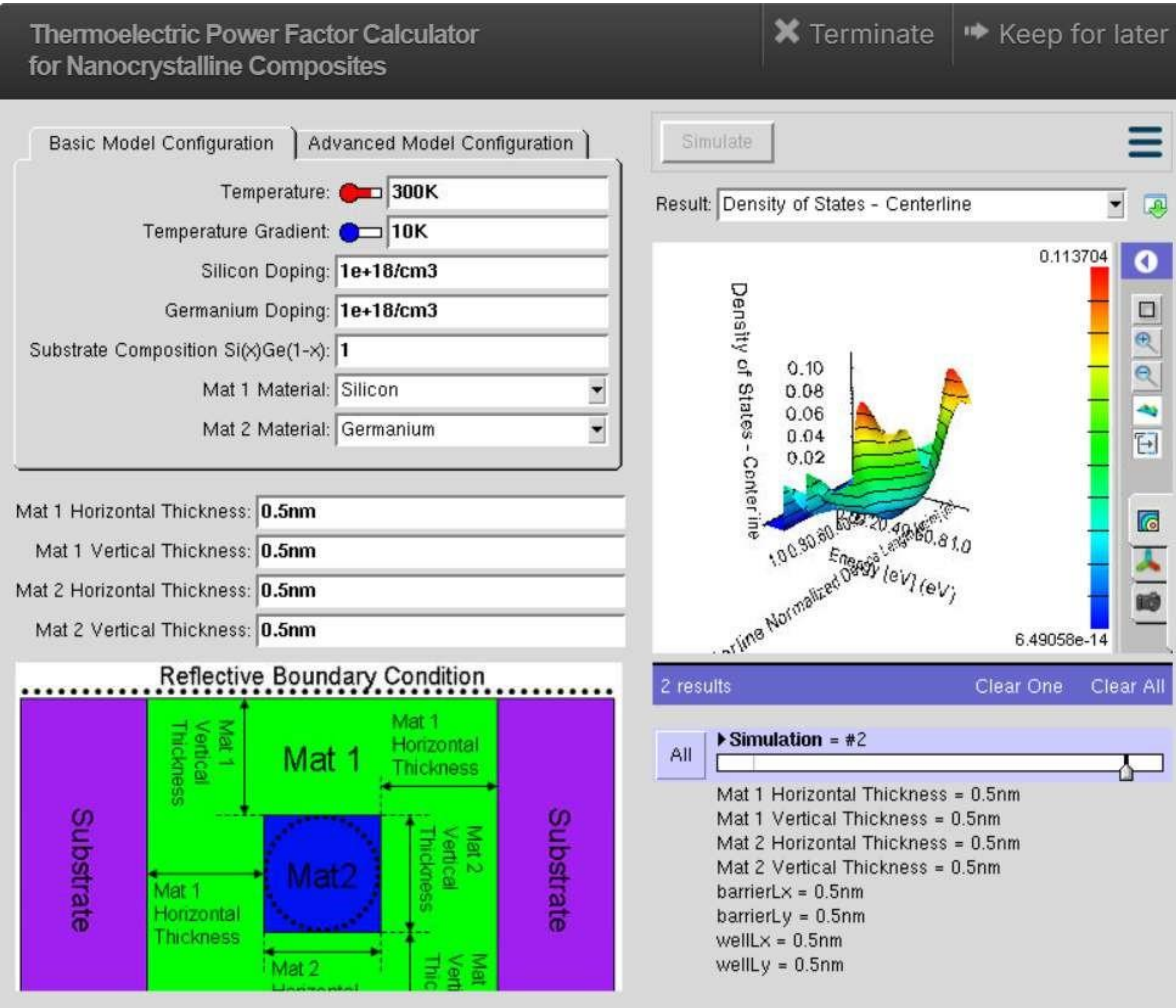
The **Density of States** tells us

How many electron energy states are available at each energy level.



Case study 1: Simulate the density of states of quantum well of thickness 0.5 nm

Aim: To simulate the **Density of States (DOS)** of a quantum well structure with a **0.5 nm well thickness** and analyze the effect of strong quantum confinement on the available electron energy states.



Case study 2: Simulate the density of states of quantum well of thickness 0.6 nm

Aim: To simulate the **Density of States (DOS)** of a quantum well structure with a **0.6 nm well thickness** and investigate the effect of increased well thickness on the electronic energy states.